

- Every value from 0 to SpeedMax SHALL be in exactly 1 range.
- Every value in any lower range shall be less than every value in any higher range.
- The value 0 (zero) in this attribute SHALL be in a range by itself. This range SHALL map to Off.
- If Auto is supported by the server, the null value SHALL be in a range by itself. This range SHALL map to Auto.
- There shall be a one-to-one mapping between each range and each value in FanMode.
- Every range SHALL map to a greater value in the FanMode attribute than all lower-valued ranges.

If a client successfully sets the FanMode attribute, the server SHALL set the SpeedSetting attribute to some value within the range which maps to the FanMode setting, as well as the PercentSetting attribute according to the equation below. If no such value exists, the attribute value SHALL NOT change.

If a client successfully sets the SpeedSetting attribute to 0, the server SHALL set the FanMode attribute to Off and the PercentSetting attribute to 0.

The following table illustrates an example of such a mapping with a value of 6 for the SpeedMax attribute. The actual mapping provided by the server is determined by the manufacturer subject to the rules mandated in this specification. Note that the values in this example are contiguous, non-overlapping and span from 0 to 6 with no missing values. Note also that each range appears in an ascending order, 0 appears by itself in a range that maps to Off, and null appears by itself in a range that maps to Auto.

Attribute	Values				
FanMode	Off	Low	Medium	High	Auto
SpeedSetting	0	1 to 2	3 to 4	5 to 6	null

The value of the PercentSetting and PercentCurrent (percent) attributes SHALL be calculate from the SpeedSetting and SpeedCurrent (speed) attributes respectively using the following formula:

- $\text{percent} = \text{floor}(\text{speed} / \text{SpeedMax} * 100)$

For example: If the SpeedMax attribute is 42 (42 speed fan) and SpeedSetting attribute is changed to 11, then PercentSetting and PercentCurrent become 26.

4.4.6.7. SpeedCurrent Attribute

This attribute SHALL indicate the actual currently operating fan speed, or zero to indicate that the fan is off. There MAY be a temporary mismatch between the value of this attribute and the value of the SpeedSetting attribute due to other system requirements or constraints that would not allow the fan to operate at the requested setting.

For example, if the value of this attribute is currently 5, and the SpeedSetting attribute is newly set to 2, the value of this attribute MAY stay above 2 for a period necessary to dissipate internal heat,